MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)

Comprehensive Refinery Safety Standard Operating Procedure (SOP) — 2026 Sovereign Edition

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1. Emergency Shutdown Procedure — Crude Distillation Unit (CDU-1, CDU-2 & CDU-3)

1.1 Trigger Conditions: Emergency Shutdown (ESD) of the Crude Distillation Unit must be initiated immediately upon: (a) Major loss of electrical power grid (>5 seconds), (b) Furnace tube rupture or unconfined crude charge leak, (c) Loss of cooling water circulation to overhead condenser batteries, or (d) Uncontrolled runaway column pressure > 45.0 bar.

1.2 Step-by-Step Shutdown Execution:

1. Trip the Emergency Shutdown Push Button (ESD-PB-01) at the Central Control Room (CCR) main console or DCS graphics.
2. Automated Emergency Block Valves (EBVs) on crude charge pumps P-101A/B will close within 3 seconds, isolating crude feed.
3. Immediately shut fuel gas and fuel oil firing valves to Atmospheric Furnaces F-101 and F-102. Inject emergency snuffing steam (15 kg/cm²) into the furnace radiant section.
4. Divert column overhead vapors to the High-Pressure Flare Header by opening flare bypass control valves FV-1044 and FV-1045.
5. Start emergency stripping steam to column bottoms to prevent heavy oil coking and vacuum collapse.
6. Establish nitrogen blanketing at 0.5 kg/cm² on column vessels and accumulator drums once system temperature drops below 200°C.

2. Hydrogen Sulfide (H2S) Toxic Gas Safety & Permissible Exposure Limits

2.1 Exposure Limits & Thresholds: Hydrogen Sulfide (H2S) is an extremely lethal, colorless gas with a rotten-egg odor at trace levels (odor fatigue occurs above 50 ppm). The mandatory MRPL exposure benchmarks comply with OISD-GDN-166 and OSHA:

- **Permissible Exposure Limit (TWA 8-hr):** 10 ppm (14 mg/m³).
- **Short Term Exposure Limit (STEL 15-min):** 15 ppm (21 mg/m³).
- **Immediately Dangerous to Life or Health (IDLH):** 100 ppm.
- **Instant Fatal Lethal Range:** > 500 ppm causes immediate pulmonary edema, neurological paralysis, and sudden cardiac arrest.

2.2 Mandatory PPE Requirements in Sour Gas & Hydrotreating Areas:

1. Every operator entering Sector-2 (DHDS, VGO-HDT, Sulfur Recovery Unit) MUST wear a continuously active 4-gas personal detector set to alarm at 5 ppm (warning) and 10 ppm (evacuate).
2. For line breaking, sampling, or maintenance where H2S > 10 ppm may be present, personnel MUST wear a Positive Pressure Self-Contained Breathing Apparatus (SCBA) with a 30-minute composite cylinder, or airline respirator with 10-minute escape bottle.
3. Air-purifying canister/cartridge respirators are strictly prohibited for H2S concentrations above 10 ppm or in confined spaces.
4. In the event of an H2S alarm, workers must immediately assemble cross-wind or up-wind at Designated Muster Point C-4.

3. Pressure Safety Valve (PSV) and Pressure Relief Valve (PRV) Compliance

3.1 Testing Frequency & Certification Schedule: All high-pressure safety relief valves (PSVs/PRVs) across MRPL must adhere to API 576 and OISD-132:

- **High-Pressure Critical Service (> 30 bar, toxic/sour):** Mandatory inspection and bench test every **12 months (1 year)**.

• **Clean Hydrocarbon Service & Atmospheric Service:** Inspection and recertification every **24 months (2 years)**.
• **Thermal Relief Valves (TRVs):** Tested during scheduled unit turnarounds (maximum 36 months).

3.2 PSV Operating and Pop Test Tolerances:
1. CDU-3 Primary relief valves (PRV-401 to PRV-408) must be set to trigger at exactly 42.5 bar gauge.
2. Set pressure tolerance: ± 3% for set pressures above 4.8 bar.
3. Cold Differential Set Pressure (CDSP) must be verified on the test bench using dry nitrogen or hydraulic test bench with certified calibrated master gauges.
4. Pop test must be performed 3 consecutive times to ensure repeatability and bubble-tight seat leakage compliance before installation.

4. Hot Work Permit Protocols in Hazardous Zones (Zone-1 & Zone-2)

4.1 Zone-1 Definition and Mandatory Prerequisites: Zone-1 is an area in which an explosive hydrocarbon atmosphere is likely to occur in normal operation. No hot work (welding, cutting, grinding, torch operation, unrated power tools) is permitted without a valid Grade-A Hot Work Permit signed by the Shift In-Charge and Safety Officer.

4.2 Step-by-Step Hot Work Execution Requirements:
1. **Gas Testing (Continuous Monitoring):** The atmosphere at the job site and surrounding 15-meter radius must be tested for Lower Explosive Limit (LEL), Oxygen, H2S, and CO.
• LEL must be **0.0%** to start hot work. Hot work is immediately cancelled if LEL exceeds **1.0%**.
• Oxygen content must be strictly between **19.5% and 23.5%**.
2. **15-Meter Radial Clearance:** All combustible materials, grease, hydrocarbons, and oily rags within 15 meters must be removed. All sewer pits, drains, and catch basins within 15 meters must be covered with fire-resistant rubber sheets and sealed with wet sand.
3. **Fire Watch & Standby Equipment:** A dedicated certified Fire Watch must be present throughout the work duration with two 10kg DCP extinguishers and a charged 2.5-inch fire hose connected to the refinery fire water ring at 8.0 kg/cm² pressure.
4. **Post-Work Watch:** Continuous fire watch must remain on site for a minimum of 30 minutes after completion of hot work to check for smoldering embers.

5. Hydrocracker Unit (HCU) & Hydrogen Generation Unit (HGU-2) Protocols

5.1 Hydrogen Hazards & High-Pressure Circulation: Hydrogen burns with an invisible flame in daylight and has a 4.0% - 75.0% explosive range in air with minimal ignition energy (0.02 mJ). Thermal imaging cameras must be deployed to detect flame fronts.
5.2 Emergency Depressurization (EDP-01): In the event of a high-pressure reactor temperature runaway (> 435°C) or major high-pressure hydrogen leak (> 140 bar), the operator must activate the Emergency Depressurization System (EDP-01). EDP-01 depressurizes the reactor loop to the flare header at a controlled rate of 7.0 bar per minute down to 20 bar, preventing catastrophic vessel embrittlement and catastrophic rupture.

6. Fire Protection, AFFF Deluge Systems & Hydrocarbon Firefighting

6.1 Fixed Deluge System Operation: All crude and LPG storage tanks are protected by medium-velocity water spray and 3% Aqueous Film-Forming Foam (AFFF) deluge systems. Minimum deluge design density is 10.2 Litres/min/m² of vessel surface area.
6.2 Fire Alarm Response Protocol: Upon activation of an automated hydrocarbon flame detector (UV/IR) or manual call point (MCP), the deluge control valve (DV-201) trips open automatically within 5 seconds. Refinery fire pump ring mains automatically maintain 10.5 kg/cm² pressure via three diesel-driven backup turbine pumps (FP-01/02/03).

7. Confined Space Entry & Positive Mechanical Isolation Protocol

7.1 Isolation and Gas Freeing: Before entering any vessel, reactor, fractionator column, or storage tank, positive mechanical isolation via spectacle blind insertion at battery limits is mandatory. Closing isolation valves alone without physical blinds or verified double-block-and-bleed is strictly prohibited under OISD-STD-105.
7.2 Entry Conditions: (a) Oxygen: strictly 19.5% - 23.5%, (b) LEL: 0.0%, (c) Toxic gases (H2S < 5 ppm, CO < 25 ppm, Benzene < 0.5 ppm, SO2 < 2 ppm). A trained standby attendant must maintain continuous verbal and visual contact at the

manhole entrance with emergency rescue winch gear.

8. Electrical Isolation & Lockout / Tagout (LOTO) Standard

All mechanical and instrument maintenance on motorized machinery (pumps, compressors, blowers, fin fans) requires an authorized LOTO certificate. The Electrical Engineer must rack out the circuit breaker at the Substation 415V/6.6kV/11kV switchgear, perform live-dead-live zero voltage verification using a calibrated proximity detector, apply an individual red padlock with a unique key, and secure the danger tag bearing the technician name, work permit number, and date. Master keys must remain locked in the personal possession of the executing engineer until completion testing.

9. Chemical Hazard Management & Neutralization Directives

9.1 Caustic Soda (NaOH 50%) & Sulfuric Acid (H2SO4 98%) Protocols: Storage areas must be surrounded by acid/alkali resistant dyke walls with a minimum 110% containment capacity of the largest storage vessel.
• For acid spills: Neutralize immediately with sodium bicarbonate or agricultural lime until pH stabilizes between 6.5 and 8.5 before transferring to the Effluent Treatment Plant (ETP).
• For caustic spills: Dilute with copiously flushed potable water and neutralize with weak acetic acid or citric acid solution.
• Emergency eyewash and deluge safety showers must be tested daily and be reachable within 10 seconds (max 15 meters) from any chemical handling point.

10. Shift Handover, Critical Alarm Log & General Safety Directives

10.1 Shift Handover Compliance: Formal shift handover must adhere to OSHA 1910.119 Process Safety Management standards. The outgoing Shift Supervisor must log: (a) All active work permits and hot work zones, (b) Bypassed or overridden ESD interlocks and safety trips, (c) Equipment running under maintenance deviation, and (d) Product quality tank line-ups.
10.2 Fall Protection & Scaffolding: 100% tie-off using a dual-lanyard full-body safety harness with shock absorber is mandatory when working at heights exceeding 1.8 meters (6 feet). All scaffolding must carry a valid Green Tag signed by a certified scaffolding inspector before access.